

ORIGINAL RESEARCH

HEART FAILURE

National Trends and Disparities in Heart Failure Hospitalizations and Inpatient Outcomes Between 2010-2022



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ABSTRACT

BACKGROUND Limited data exist on trends and differential heart failure (HF) hospitalization rates, particularly when appropriate statistical age standardization is applied.

OBJECTIVES This study aimed to assess temporal trends in age-standardized hospitalization rates and in-hospital mortality for primary HF hospitalizations among younger (<65 years) and older (>64 years) patients, stratified by sex, race/ethnicity, insurance status, and urban/rural hospital location.

METHODS The National Inpatient Sample database was analyzed to identify adults who were hospitalized for HF between 2010 and 2022. HF hospitalization rates and in-hospital mortality trends were then examined across demographic strata.

RESULTS In total, 14,287,733 weighted hospitalizations for HF were identified. Overall, age-standardized HF hospitalizations per 100,000 U.S. population showed a nonsignificant change from 406 in 2010 to 381 in 2014 ($P = 0.14$), followed by an increase to 447 in 2018 ($P = 0.002$) and a subsequent decline to 395 in 2022 ($P = 0.002$). Hospitalization rates declined at rural hospitals but increased among Medicare and Medicaid insurers ($P < 0.05$). From 2010 to 2022, the <65-year age group experienced increasing hospitalizations (from 124 to 161; average annual percentage change: 2.2 [95% CI: 1.6-2.8]; $P < 0.001$) in contrast to the >64-year age group (from 1,775 to 1,525; average annual percentage change: -1.2; 95% CI: -2.8 to 0.9). In-hospital mortality for the overall cohort did not improve between 2010 and 2022, and with a significant uptrend observed in 2020 compared with the 2010-2019 period ($P = 0.001$).

CONCLUSIONS This longitudinal analysis of U.S. hospitalizations from 2010 to 2022 revealed that HF hospitalizations began increasing in 2014, peaked in 2019, and then declined sharply in 2020, a trend that persisted through 2022, with a more pronounced decline among adults aged >64 years. In contrast, HF hospitalizations among adults aged <65 years continued to rise, highlighting a shift in epidemiological landscape in which HF increasingly affects younger populations. In-hospital mortality increased in 2020, changing the trend observed in prior years. (JACC Heart Fail. 2026;14:102777) © 2026 by the American College of Cardiology Foundation.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**ABBREVIATIONS
AND ACRONYMS****AAPC** = average annual
percentage change**APC** = annual percentage
change**CV** = cardiovascular**HF** = heart failure**NH** = non-Hispanic

Hear failure (HF) is the leading cause of hospitalization for cardiovascular (CV) conditions in the United States, with a high risk for readmissions.¹ HF hospitalizations are a major driver of morbidity, mortality, and health care expenditures.²⁻⁴ Recent data describing national temporal trends in age-standardized rates for HF hospitalization based on sex, race/ethnicity, insurance status, and urban/rural hospital location status are lacking.⁵⁻⁸ Limited data exist on trends and differential HF hospitalization rates, particularly when appropriate statistical age standardization is applied.⁵ Such demographically standardized hospitalization rates are useful for assessing temporal trends and differences in HF hospitalization and mortality burden. Hence, we examined national trends in age-standardized HF hospitalizations and in-hospital mortality rates from 2010 to 2022 in the United States, stratified by sex, race/ethnicity, insurance payer type, and hospital urban/rural location (reported separately for age groups <65 years and ≥65 years).

METHODS

We used the 2010-2022 NIS (National Inpatient Sample), a database maintained as part of the HCUP (Healthcare Cost and Utilization Project) sponsored by the Agency for Healthcare Research and Quality.⁹ The NIS, an all-payer database, provides accurate estimates representing >97% of the U.S. population by using validated survey-weighting methodology.⁹ This study was deemed exempt from institutional review board approval, given the de-identified nature of the data and the availability of the database to the public. For each year of the NIS, the number of adult HF hospitalizations (using International Classification of Diseases-9th Revision and International Classification of Diseases-10th Revision codes)¹ was estimated for the overall cohort and by subgroups of sex, race/ethnicity (non-Hispanic [NH]-White, NH-Black, Hispanic, and NH-Asian and Pacific Islander), primary insurance payer type (Medicare, Medicaid, and Private), and hospital location (urban/rural). A primary HF hospitalization was defined as one with an HF discharge code as the primary designated diagnostic code. We estimated crude HF hospitalization rates per 100,000 U.S. adults using U.S. Census Bureau estimates for each subgroup and the overall cohort across each study year and age-standardized for the 2000 U.S. standard population using the direct standardization method, as reported

previously.⁵ For subgroup analyses, respective U.S. subpopulation census estimates were used. The rates were calculated for the overall group and by age groups—younger (<65 years) and older (≥65 years)—across different demographic strata. We performed single year of life adjustments (for race/ethnic subgroups and the overall cohort) and each decade of life for different insurance and urban/rural location subgroups, as available in U.S. census data files. The in-hospital mortality for each group was standardized to the 2000 U.S. standard population using the direct standardization method and reported as proportions. For all data sets, missing race/ethnicity was imputed using a multinomial logistic model incorporating age, sex, insurance status, comorbid conditions, hospital region, and characteristics, as done previously.⁵ The primary purpose of the imputation was to normalize population-based estimates and not to reliably identify the missing group for any single HF hospitalization. Of note, for the age-based U.S. population, insurance-based census files were not available for the year 2021.

Trends in age-adjusted hospitalizations and inpatient mortality were examined using the Joinpoint Regression Program. This program identifies significant changes in annual trends through joinpoint regression, which fits models of linear segments where temporal variation occurred. Annual percentage change (APC) with 95% CI among intervals identified using joinpoint regression and average annual percentage change (AAPC) with 95% CI were calculated for the overall 2010-2022 period. Joinpoint regression was performed using Joinpoint version 5.4.0.0 (available from the National Cancer Institute). All other analyses were performed using Stata 18.1 (StataCorp LLC), and all estimations were derived with the appropriate NIS survey weights. To compare the changes in rates between groups and age-based subgroups (race/ethnicity, insurance, sex, and urban vs rural) during time periods, regression analyses were performed with time period interaction wherever applicable (see outcomes and covariates described under each model's results in the [Supplemental Tables 1 to 9](#)). The assumptions of normality and homoscedasticity were verified through visual inspection of residual plots. We also performed interrupted time series analysis using the year 2020 to estimate the effect of COVID-19 on trends.

RESULTS

AGE-STANDARDIZED PRIMARY HF HOSPITALIZATION RATE TRENDS. The absolute number of HF hospitalizations changed from 997,244 in 2010 to 1,213,844 in

2022 (Table 1). The age-standardized HF hospitalization rates (per 100,000 U.S. adults) for the overall cohort changed negligibly from 406 in 2010 to 395 in 2022 (AAPC: -0.5 [95% CI: -1.3 to 0.3]; $P = 0.18$). The rates increased between 2014 and 2018 (APC: $+4.3$ [95% CI: 1.9 - 8.3]; $P = 0.002$) followed by a decrease from 2018 onward (APC: -3.6 [95% CI: -7.8 to -1.6]; $P = 0.002$) (Figure 1, Supplemental Table 1).

Among individuals aged ≥ 65 years, changes in rates were nonsignificant: 1,775 in 2010 to 1,525 in 2022 (AAPC: -1.2 [95% CI: -2.8 to 0.9]; $P = 0.15$). Among adults aged < 65 years, hospitalization rates increased from 124 in 2010 to 161 in 2022 (AAPC: $+2.2$ [95% CI: 1.6 - 2.8]; $P < 0.001$) (Figure 1, Central Illustration). In subgroup analysis within the < 65 -year age group, the proportion of patients in the middle-age group (35-65 years) increased (Supplemental Figure 1) For males (AAPC: -0.4 [95% CI: -1.2 to 0.3]; $P = 0.24$) and females (AAPC: $+0.1$ [95% CI: -1.1 to 1.4]; $P = 0.83$), there was no significant change in rates from 2010 to 2022 (Table 2).

The rate did not change significantly among different racial/ethnic subgroups from 2010 to 2022, except for Blacks (AAPC: -1.1 [95% CI: -1.9 to -0.2]; $P = 0.01$) (Figure 2, Table 3). Subgroup analysis by hospital location demonstrated a nonsignificant change in rates in urban locations between 2010 and 2022 (AAPC: -0.2 [95% CI: -1.1 to 0.7]; $P = 0.50$). Rural hospitals had a significant decline from 249 in 2010 to 176 in 2022 (AAPC: -3.0 [95% CI: -4.3 to -1.8]; $P < 0.001$) (Table 4). Trends stratified by insurance type revealed an increase in hospitalization rates for Medicare (AAPC: $+0.8$ [95% CI: 0.1 - 1.4]; $P = 0.02$) and Medicaid (AAPC: $+1.7$ [95% CI: 1.0 - 2.4]; $P < 0.001$) insurers (Table 4).

On analyzing the different subgroups based on age groups, we found that among the older age group (≥ 65 years), hospitalization rates from 2010 to 2022 declined across the following subgroups: males (AAPC: -1.7 [95% CI: -2.6 to -0.9]; $P < 0.001$), Blacks (AAPC: -2.5 [95% CI: -3.5 to -1.5]; $P < 0.001$), Hispanics (AAPC: -2.2 [95% CI: -3.8 to -0.5]; $P = 0.009$), Medicare insurers (AAPC: -1.6 [95% CI: -2.3 to -1.1]; $P < 0.001$), and rural hospital locations (AAPC: -4.4 [95% CI: -5.6 to -3.2]; $P < 0.001$) across the study period (Tables 2 to 4). The rates increased for all subgroups among the younger (< 65 -year) age group across the study period ($P < 0.05$ for all), except for Medicaid insurers, Blacks, and Hispanics (Tables 3 and 4).

From 2010 to 2022, the younger age group had a significant increase in rates compared with the older age group (Supplemental Table 2). There was a

greater decline in HF hospitalization rates for older patients compared with the younger age group ($P < 0.001$) in the post-COVID-19 pandemic year (2020) (Supplemental Table 3). There were no significant differences in rates between males and females (Supplemental Table 4). HF hospitalization rates declined in rural locations compared with urban locations ($P < 0.001$) (Supplemental Table 5). Within insurance groups, Medicaid insurers had a more significant increase in rates than did private insurers ($P = 0.032$) (Supplemental Table 6). Younger Medicaid- and Medicare-insured groups had higher hospitalization rates than did the older age group (Supplemental Table 7).

AGE-STANDARDIZED IN-HOSPITAL MORTALITY TRENDS. Age-standardized in-hospital mortality rates showed a nonsignificant change from 1.8% in 2010 to 1.8% in 2022 ($P = 0.97$). There was a downward trend from 1.8% in 2010 to 1.6% in 2019, but this trend was not statistically significant ($P = 0.28$) (Supplemental Table 8). However, rates in the post-COVID-19 pandemic year (2020) increased significantly compared with the 2010-2019 period ($P = 0.004$) (Figure 3A, Supplemental Table 8).

The in-hospital mortality for the older age group had a declining trend (3.3% in 2010 to 2.8% in 2019; APC: -1.7 [95% CI: -3.9 to -0.8]; $P < 0.001$), followed by an increase from 2.8% in 2019 to 3.2% in 2022 (APC: 4.2 [95% CI: 0.6 - 9.1]; $P = 0.02$). (Figure 3, Supplemental Table 9). Among the younger age group, there was no significant change in age-standardized in-hospital mortality (1.5% in 2010 to 1.6% in 2022; $P = 0.99$). Rates for different subgroups are listed in Table 5.

DISCUSSION

Using a large U.S. administrative database, we examined age-standardized HF hospitalization rates and in-hospital mortality across the 2 age groups (< 65 and ≥ 65 years), stratified by sex, race/ethnicity, insurance payer systems, and urban/rural hospital location. In this nationally representative cohort, we observed a steady increase in HF hospitalization from 2014 to 2019 for the overall cohort, suggesting that declining HF hospitalization trends in the decade before 2014 were disrupted.^{5,6,10} The increasing hospitalization rates likely correlate with a higher prevalence of HF risk factors, greater comorbidities, and lack of access to guideline-directed medical therapies.^{4,11} Notably, this period was followed by a sharp decline in HF hospitalization, consistent with previously reported trends in hospitalizations for high-acuity CV conditions during the COVID-19

TABLE 1 Baseline Characteristics for the National HF Cohort From 2010 to 2022 From the National Inpatient Sample

	2010 (n = 997,244)	2011 (n = 1,003,419)	2012 (n = 951,220)	2013 (n = 960,124)	2014 (n = 989,080)	2015 (n = 1,047,865)
Patient characteristics						
Age, y	72.6 (72.1-73.1)	73.0 (72.6-73.5)	72.5 (72.3-72.8)	72.3 (72.0-72.5)	72.0 (71.8-72.2)	71.9 (71.7-72.1)
Female, %	49.9 (49.4-50.4)	50.5 (50.0-50.9)	49.9 (49.5-50.2)	49.0 (48.7-49.4)	48.6 (48.3-48.9)	48.4 (48.1-48.7)
Race, %						
NH-White	65.9 (62.9-8.8)	67.1 (64.6-69.5)	67.3 (66.1-8.6)	67.5 (66.2-8.7)	67.1 (65.9-8.3)	66.2 (64.9-7.4)
NH-Black	21.8 (19.3-24.5)	20.8 (18.8-23.0)	20.2 (19.2-21.3)	20.2 (19.1-21.2)	20.6 (19.6-21.7)	21.4 (20.4-22.5)
Hispanic	7.7 (6.3-9.4)	7.8 (6.4-9.5)	7.4 (6.8-8.1)	7.7 (7.1-8.4)	7.5 (6.9-8.2)	7.7 (7.1-8.3)
NH- Asian and PI	1.9 (1.5-2.4)	1.6 (1.3-1.9)	1.9 (1.7-2.2)	2.0 (1.8-2.3)	2.0 (1.7-2.2)	2.2 (1.9-2.5)
Native American	0.8 (0.5-1.2)	0.5 (0.3-0.8)	0.6 (0.4-0.7)	0.5 (0.4-0.6)	0.5 (0.4-0.6)	0.5 (0.4-0.5)
Insurance, %						
Medicare	73.4 (71.9-4.8)	75.8 (74.7-76.9)	75.9 (75.4-6.5)	75.0 (74.4-5.6)	74.3 (73.7-4.9)	74.1 (73.5-4.7)
Medicaid	8.5 (7.7-9.4)	7.6 (7.0-8.3)	8.0 (7.7-8.4)	8.1 (7.7-8.5)	9.4 (9.0-9.8)	9.8 (9.4-10.2)
Private	12.7 (11.8-3.6)	11.6 (10.9-2.3)	10.7 (10.4-1.1)	11.1 (10.7-1.5)	11.6 (11.2-2.0)	11.6 (11.2-2.0)
Median household income (in quartiles), %						
0-25th percentile	33.2 (30.7-35.7)	32.9 (30.5-5.3)	34.6 (33.3-5.9)	32.8 (31.6-4.1)	33.2 (32.0-4.5)	34.6 (33.5-5.8)
26th-50th percentile	26.4 (24.6-28.2)	25.3 (23.7-6.9)	25.3 (24.5-6.1)	26.8 (26.0-7.6)	28.1 (27.2-8.9)	24.9 (24.1-5.7)
51st-75th percentile	22.7 (21.1-24.3)	24.2 (22.6-5.8)	22.3 (21.6-3.1)	22.7 (21.9-3.5)	21.6 (20.8-2.3)	22.8 (22.1-3.5)
76th-100th percentile	17.8 (15.7-20.2)	17.7 (15.7-9.8)	17.8 (16.7-8.8)	17.7 (16.6-8.8)	17.1 (16.1-8.2)	17.7 (16.7-8.7)
Comorbidities, %						
AMI	3.5 (3.3-3.8)	3.4 (3.2-3.6)	3.6 (3.5-3.7)	3.6 (3.5-3.8)	3.6 (3.5-3.8)	3.6 (3.4-3.7)
Anemia	31.3 (31.3-31.3)	34.1 (34.1-34.1)	34.3 (34.3-34.3)	34.3 (34.3-34.3)	34.6 (34.6-34.6)	34.2 (34.2-34.2)
AF	36.3 (35.3-37.2)	38.6 (37.8-39.4)	39.5 (39.1-40.0)	40.5 (40.1-41.0)	41.4 (40.9-41.8)	43.1 (42.6-43.5)
COPD	29.2 (28.4-30.0)	30.5 (29.7-31.3)	30.9 (30.5-31.3)	31.0 (30.6-31.4)	31.1 (30.7-31.6)	32.0 (31.6-32.4)
CAD	53.7 (52.7-54.8)	54.7 (53.8-55.7)	55.7 (55.2-56.1)	55.3 (54.9-55.8)	55.3 (54.8-55.7)	55.2 (54.7-55.6)
Diabetes	44.5 (43.7-45.3)	45.7 (45.0-46.4)	46.6 (46.2-46.9)	47.4 (47.0-47.7)	47.9 (47.6-48.2)	48.2 (47.9-48.5)
Fluid and electrolyte disorders	27.3 (26.5-28.1)	29.3 (28.5-30.1)	30.6 (30.2-31.0)	32.0 (31.6-32.4)	33.6 (33.2-34.0)	34.2 (33.8-34.6)
HTN	75.7 (74.7-76.6)	77.5 (76.7-78.3)	79.6 (79.3-80.0)	80.3 (79.9-80.7)	81.8 (81.5-82.2)	83.7 (83.3-84.0)
Obesity	14.2 (13.6-14.7)	16.5 (16.0-17.0)	17.9 (17.6-18.2)	19.2 (18.9-19.5)	20.9 (20.6-21.2)	22.2 (21.9-22.5)
PVD	7.7 (7.3-8.1)	8.3 (7.9-8.7)	8.5 (8.3-8.7)	8.5 (8.3-8.7)	8.7 (8.5-8.8)	9.5 (9.3-9.7)
Valvular disease	26.8 (25.7-27.9)	28.6 (27.6-9.7)	29.1 (28.6-29.7)	29.3 (28.8-29.8)	29.9 (29.4-30.5)	30.0 (29.5-30.5)
VT	4.9 (4.6-5.3)	5.1 (4.8-5.4)	5.2 (5.0-5.4)	5.2 (5.0-5.4)	5.3 (5.1-5.5)	5.4 (5.2-5.6)
Hospital characteristics, %						
Hospital bed size						
Small	13.9 (12.1-15.9)	14.0 (12.2-16.0)	15.4 (14.3-16.6)	15.2 (14.0-16.4)	20.1 (18.8-21.6)	19.5 (18.2-20.9)
Medium	23.3 (20.4-26.4)	24.1 (21.2-27.3)	26.9 (25.1-28.7)	27.0 (25.2-28.9)	29.8 (27.9-31.7)	30.1 (28.2-32.0)
Large	62.9 (59.3-66.3)	61.9 (58.3-65.4)	57.7 (55.6-59.8)	57.8 (55.7-59.9)	50.1 (47.8-52.3)	50.4 (48.2-52.6)
Hospital region						
Northeast	21.1 (18.0-24.6)	20.7 (17.5-24.2)	20.2 (18.4-22.1)	20.0 (18.2-21.9)	19.8 (18.0-21.7)	19.4 (17.6-21.2)
Midwest or North Central	23.1 (20.1-26.4)	23.0 (20.1-26.1)	23.2 (21.5-25.1)	22.9 (21.2-24.8)	23.3 (21.5-25.1)	23.0 (21.2-24.8)
South	40.6 (36.6-44.7)	40.5 (36.5-44.6)	40.8 (38.6-43.1)	40.9 (38.6-43.2)	41.1 (38.9-43.4)	41.4 (39.2-43.7)
West	15.2 (12.7-18.0)	15.9 (13.4-18.8)	15.7 (14.4-17.2)	16.1 (14.8-17.6)	15.8 (14.5-17.3)	16.2 (14.8-17.7)
Location						
Urban	85.3 (83.0-87.3)	85.4 (83.1-87.4)	85.6 (84.4-86.7)	85.8 (84.7-86.9)	87.7 (86.7-88.7)	88.2 (87.2-89.1)
Rural	14.7 (12.7-17.0)	14.6 (12.6-16.9)	14.4 (13.3-15.6)	14.2 (13.1-15.3)	12.3 (11.3-13.3)	11.8 (10.9-12.8)
Outcomes						
Length of stay, d	5.2 (5.1-5.3)	5.1 (5.0-5.3)	5.2 (5.2-5.3)	5.3 (5.2-5.4)	5.3 (5.2-5.4)	5.3 (5.3-5.4)
Age-unadjusted in-hospital mortality	3.1 (2.9-3.2)	3.1 (2.9-3.2)	3.1 (3.0-3.2)	3.0 (2.9-3.1)	2.9 (2.8-3.0)	2.9 (2.8-3.0)

Values are HR (95% CI).

AF = atrial fibrillation; AMI = acute myocardial infarction; CAD = coronary artery disease; COPD = chronic obstructive pulmonary disease; HF = heart failure; HTN = hypertension; NH = non-Hispanic; PI = Pacific Islander; PVD = peripheral vascular disease; VT = ventricular tachycardia.

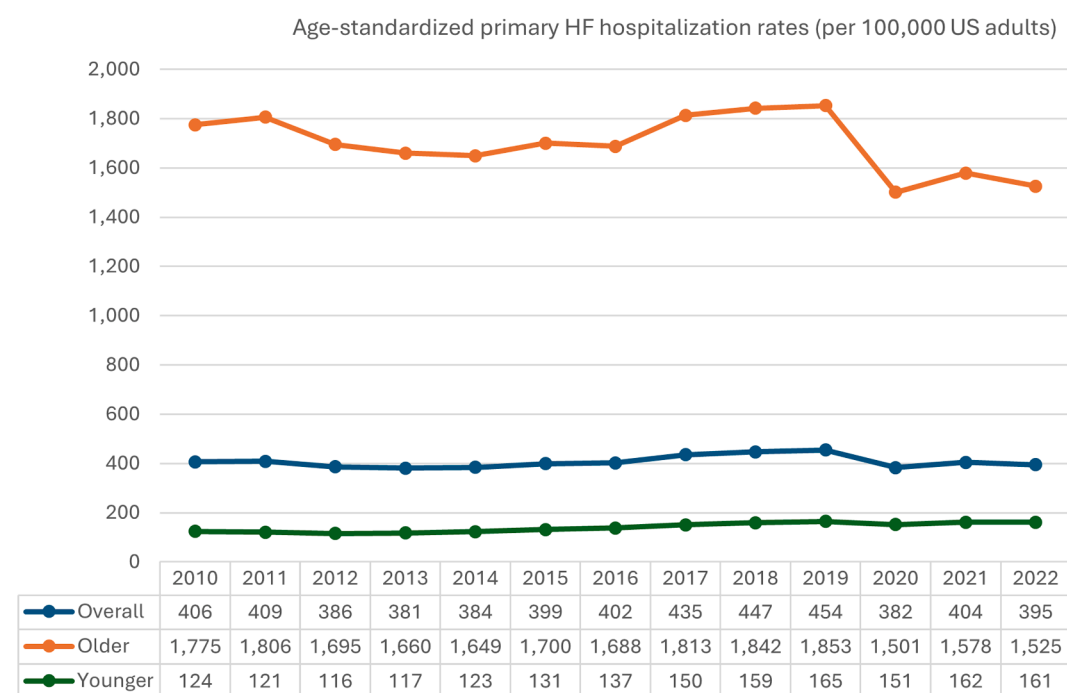
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TABLE 1 Continued

2016 (n = 1,081,154)	2017 (n = 1,191,355)	2018 (n = 1,248,490)	2019 (n = 1,295,135)	2020 (n = 1,109,965)	2021 (n = 1,198,838)	2022 (n = 1,213,844)
71.6 (71.4-1.8) 48.3 (48.0-8.6)	71.6 (71.4-1.8) 48.1 (47.8-48.4)	71.4 (71.2-1.6) 47.6 (47.3-47.9)	71.3 (71.1-1.5) 47.5 (47.2-47.8)	70.5 (70.3-0.7) 46.5 (46.3-46.8)	70.6 (70.4-0.8) 46.7 (46.5-47.0)	70.7 (70.7-70.8) 46.7 (46.5-46.9)
66.0 (64.8-7.2) 21.3 (20.3-22.3) 8.0 (7.4-8.6) 2.1 (1.9-2.4) 0.5 (0.4-0.6)	65.1 (63.8-6.3) 21.6 (20.5-22.6) 8.4 (7.8-9.0) 2.2 (1.9-2.5) 0.5 (0.4-0.6)	64.8 (63.5-6.0) 21.2 (20.2-22.3) 9.0 (8.3-9.7) 2.2 (2.0-2.5) 0.5 (0.5-0.6)	65.1 (63.9-6.3) 21.6 (20.6-22.7) 8.2 (7.7-8.9) 2.2 (2.0-2.5) 0.6 (0.5-0.7)	64.6 (63.4-5.8) 22.1 (21.0-23.1) 8.3 (7.7-8.9) 2.1 (1.9-2.4) 0.6 (0.5-0.7)	64.5 (63.3-5.6) 21.7 (20.7-22.7) 8.8 (8.2-9.5) 2.3 (2.0-2.5) 0.6 (0.5-0.7)	63.4 (63.3-3.7) 22.1 (21.9-22.2) 9.2 (9.1-9.3) 2.3 (2.3-2.4) 0.6 (0.6-0.6)
73.7 (73.1-4.3) 10.3 (9.9-10.8) 11.3 (10.9-1.7)	74.1 (73.6-4.7) 10.4 (10.0-0.8) 10.9 (10.5-1.2)	73.6 (73.0-4.2) 10.6 (10.2-1.1) 11.0 (10.6-1.4)	72.7 (72.1-3.3) 11.1 (10.6-1.5) 11.2 (10.8-1.6)	70.3 (69.7-0.9) 12.1 (11.7-2.6) 11.9 (11.5-2.4)	69.7 (69.1-0.3) 12.8 (12.3-3.2) 11.9 (11.4-2.4)	69.4 (69.2-9.6) 12.9 (12.7-3.0) 12.1 (11.9-2.2)
34.6 (33.4-5.7) 25.8 (25.1-6.6) 22.6 (21.9-3.4) 17.0 (16.1-8.0)	34.2 (33.1-5.4) 26.8 (26.1-7.6) 22.1 (21.4-2.8) 16.8 (15.9-7.8)	33.2 (32.0-4.4) 27.4 (26.6-8.2) 22.6 (21.9-3.4) 16.8 (15.9-7.8)	34.1 (32.9-5.3) 25.8 (25.0-6.5) 22.9 (22.2-3.7) 17.2 (16.2-8.2)	34.4 (33.2-5.6) 27.8 (27.0-8.6) 21.4 (20.7-2.1) 16.4 (15.5-7.4)	33.8 (32.6-5.0) 25.8 (25.0-6.5) 22.7 (22.0-3.5) 17.7 (16.7-8.8)	33.1 (32.9-3.3) 26.3 (26.1-6.5) 23.1 (22.9-3.2) 17.5 (17.4-7.7)
3.6 (3.4-3.8) 33.6 (33.6-33.6) 45.0 (44.6-45.5) 35.4 (34.9-35.8) 54.7 (54.2-55.1) 47.4 (47.0-47.8) 34.9 (34.5-35.3) 86.7 (86.4 87.0) 24.7 (24.3-25.0) 11.0 (10.8-11.3) 29.5 (29.0-30.0) 5.1 (5.0-5.3)	4.3 (4.1-4.5) 34.3 (34.3-34.3) 45.8 (45.3-46.2) 35.8 (35.3-36.2) 54.6 (54.2-55.1) 48.7 (48.4-49.0) 36.5 (36.2-36.9) 91.2 (91.0-91.4) 25.8 (25.5-26.2) 9.0 (8.8-9.3) 28.9 (28.4-29.4) 5.2 (5.0-5.4)	5.6 (5.3-5.9) 34.4 (34.4-34.4) 46.5 (46.1-47.0) 35.8 (35.4-36.3) 54.5 (54.0-54.9) 48.8 (48.4-49.1) 37.9 (37.5-38.3) 92.5 (92.3-92.8) 27.1 (26.7-27.5) 7.7 (7.4-7.9) 29.0 (28.5-29.5) 5.4 (5.2-5.6)	6.6 (6.3-6.9) 35.1 (35.1-35.1) 47.0 (46.5-47.4) 35.7 (35.3-36.1) 54.6 (54.1-55.0) 48.9 (48.6-49.3) 38.7 (38.3-39.1) 93.2 (93.0-93.4) 27.7 (27.3-28.1) 7.6 (7.3-7.9) 29.4 (28.9-29.9) 5.9 (5.7-6.1)	8.6 (8.3-9.0) 36.1 (36.1-36.1) 46.7 (46.3-47.2) 34.8 (34.4-35.3) 54.5 (54.0-54.9) 49.5 (49.1-49.8) 42.0 (41.6-42.4) 93.3 (93.1-93.5) 30.0 (29.6-30.5) 7.5 (7.2-7.8) 29.0 (28.5-29.5) 6.6 (6.4-6.8)	9.7 (9.3-10.1) 36.4 (36.4-36.4) 46.9 (46.5-47.4) 33.6 (33.2-34.0) 53.5 (53.0-54.0) 48.7 (48.3-49.0) 42.1 (41.7-42.5) 93.6 (93.4-93.8) 31.0 (30.6-31.4) 7.5 (7.2-7.7) 29.5 (29.1-30.0) 6.6 (6.4-6.8)	11.0 (10.8-11.1) 37.4 (37.2-37.6) 47.5 (47.3-47.7) 32.8 (32.6-33.0) 53.1 (52.9-53.3) 48.2 (48.0-48.4) 42.7 (42.5-42.9) 93.8 (93.7-93.9) 30.8 (30.6-31.0) 7.3 (7.2-7.4) 30.7 (30.5-30.9) 5.0 (4.9-5.1)
20.1 (18.7-21.5) 29.3 (27.4-31.2) 50.6 (48.4-52.8)	21.1 (19.7-22.6) 30.6 (28.6-32.5) 48.4 (46.1-50.6)	22.4 (20.9-23.9) 30.0 (28.1-32.0) 47.6 (45.3-49.9)	23.6 (22.1-25.2) 29.7 (27.8-31.7) 46.6 (44.4-48.9)	24.1 (22.6-25.8) 28.8 (26.9-30.8) 47.0 (44.8-49.3)	24.7 (23.1-26.3) 28.7 (26.8-30.6) 46.6 (44.3-48.9)	24.9 (24.8-25.1) 29.4 (29.3-29.6) 45.6 (45.4-45.8)
19.0 (17.3-20.8) 23.0 (21.2-24.8) 41.4 (39.2-43.6) 16.7 (15.3-18.2)	18.7 (17.0-20.6) 23.0 (21.2-24.8) 41.2 (39.0-43.5) 17.1 (15.6-18.6)	18.5 (16.8-20.4) 23.1 (21.3-25.0) 41.1 (38.9-43.3) 17.3 (15.8-18.8)	18.3 (16.6-20.1) 23.0 (21.2-24.9) 41.2 (39.0-43.5) 17.5 (16.0-19.1)	17.8 (16.1-19.6) 22.8 (21.0-24.8) 41.9 (39.6-44.1) 17.5 (16.0-19.1)	18.4 (16.7-20.2) 22.9 (21.1-24.8) 41.4 (39.1-43.6) 17.4 (15.8-19.0)	17.8 (17.7-18.0) 22.4 (22.2-22.6) 42.0 (41.8-42.2) 17.8 (17.6-17.9)
88.6 (87.7-89.5) 11.4 (10.5-12.3)	89.0 (88.1-89.9) 11.0 (10.1-11.9)	89.5 (88.5-90.4) 10.5 (9.6-11.5)	89.7 (88.8-90.6) 10.3 (9.4-11.2)	89.7 (88.8-90.6) 10.3 (9.4-11.2)	90.2 (89.3-91.0) 9.8 (9.0-10.7)	90.1 (90.0-90.3) 9.9 (9.7-10.0)
5.3 (5.3-5.4) 2.8 (2.7-2.9)	5.4 (5.3-5.4) 2.8 (2.7-2.8)	5.4 (5.3-5.5) 2.6 (2.5-2.7)	5.5 (5.4-5.5) 2.5 (2.4-2.6)	5.6 (5.5-5.7) 2.8 (2.7-2.8)	5.7 (5.7-5.8) 2.9 (2.8-3.0)	5.9 (5.9-6.0) 2.9 (2.9-3.0)

pandemic. The magnitude of this decline was more pronounced among the older age group (≥ 65 years) than among the younger age group (< 65 years), potentially driven by previously reported factors, including lack of patient presentation, increased morbidity and mortality outside the hospital, COVID-

19-related deaths among HF patients who are at risk for high-acuity CV events, HF management at home, and reductions in adverse events.¹² Although we are unable to fully study the mechanisms behind the decline among older-age HF hospitalizations from 2019 onward, it may be partly due to survivorship

FIGURE 1 Age-Standardized Primary HF Hospitalization Rates (per 100,000 U.S. Adults)

Age-standardized primary HF hospitalization rates (per 100,000 U.S. adults) from 2010 to 2022 for the overall cohort and by age group: younger (<65 years) and older (≥65 years). HF = heart failure.

bias, as there was a large rise in population deaths during the pandemic, especially among high-risk patients in the community who were outside the medical system.^{12,13} This could have been due to COVID-19-related deaths in the community that may have occurred disproportionately among high-risk adults, such as older HF patients. Notably, this decline in HF hospitalizations should not be interpreted as a positive trend, as it may reflect deterioration in CV care progress.¹³ Although HF hospitalization rates among the older age group partially rebounded since 2020, they remained lower than pre-pandemic levels (2010-2019), consistent with prior regional studies.¹² Further studies are needed to identify and quantify such factors.

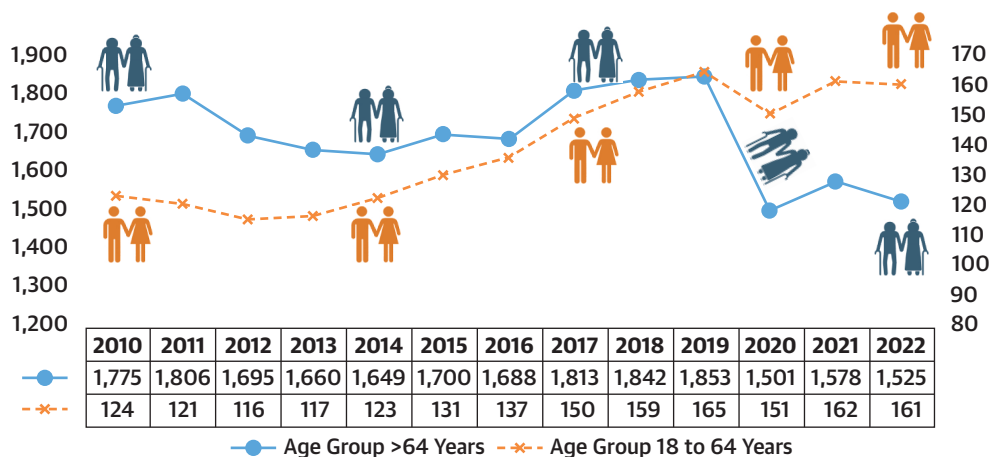
Compared to trends among older (≥65 years) adults, HF hospitalization rates among younger (<65 years) individuals increased from 2010 to 2019, followed by a similar, albeit less pronounced, drop in 2020 and a subsequent partial rebound in 2021 and 2022. However, in contrast to older adults, hospitalization rates among the younger group remained higher in 2021 and 2022 than in the 2010-2018 period, highlighting a divergent pattern between the 2 age

groups with a marked burden of HF hospitalization among younger populations. The increased HF hospitalization rates among the younger age group was observed across White and Asian racial subgroups. In addition, HF hospitalization rates among younger Medicare beneficiaries increased substantially from 2010 to 2022, whereas rates among their older counterparts declined during this period. These findings suggest that HF is now affecting younger populations and is not limited to the aging population.⁴ Although the exact reasons for this phenomenon remain unclear, plausible explanations include increasing comorbidity burden, limited access to health care services, and guideline-directed therapies, signifying a shift in the epidemiological landscape of HF.¹⁴

Our findings revealed no significant reduction in age-standardized HF hospitalization rates across overall racial/ethnic groups, except for NH-Blacks. NH-Black adults consistently had the highest HF hospitalization rates each year from 2010 to 2022. Prior studies have reported similar disparities,¹⁵ attributing them to a higher burden of cardiometabolic comorbidities, socioeconomic barriers to health care access, and structural inequities in

CENTRAL ILLUSTRATION National Trends and Disparities in Heart Failure Hospitalizations

Age-Standardized HF Hospitalization Rates (per 100,000 U.S. Adults) for Older Age Groups (Primary Y-Axis) and Younger Age Groups (Secondary Y-Axis)



• NH-Black adults had the highest HF hospitalization rates each year from 2010 to 2022 with an overall declining trend.



• Uptrend in hospitalization rates for Medicare and Medicaid insurers. Medicaid beneficiaries experienced significant increase compared to private insurers.



• Lower hospitalization rates in rural hospitals compared to urban centers with an overall decrease in rural HF hospitalizations.



• Age-standardized in-hospital mortality had an increase in 2020 disrupting prior year trends, coinciding with the onset of the COVID-19 pandemic.

Agarwal MA, et al. JACC Heart Fail. 2026;14(4):102777.

HF = heart failure; NH = non-Hispanic. Created with BioRender.com.

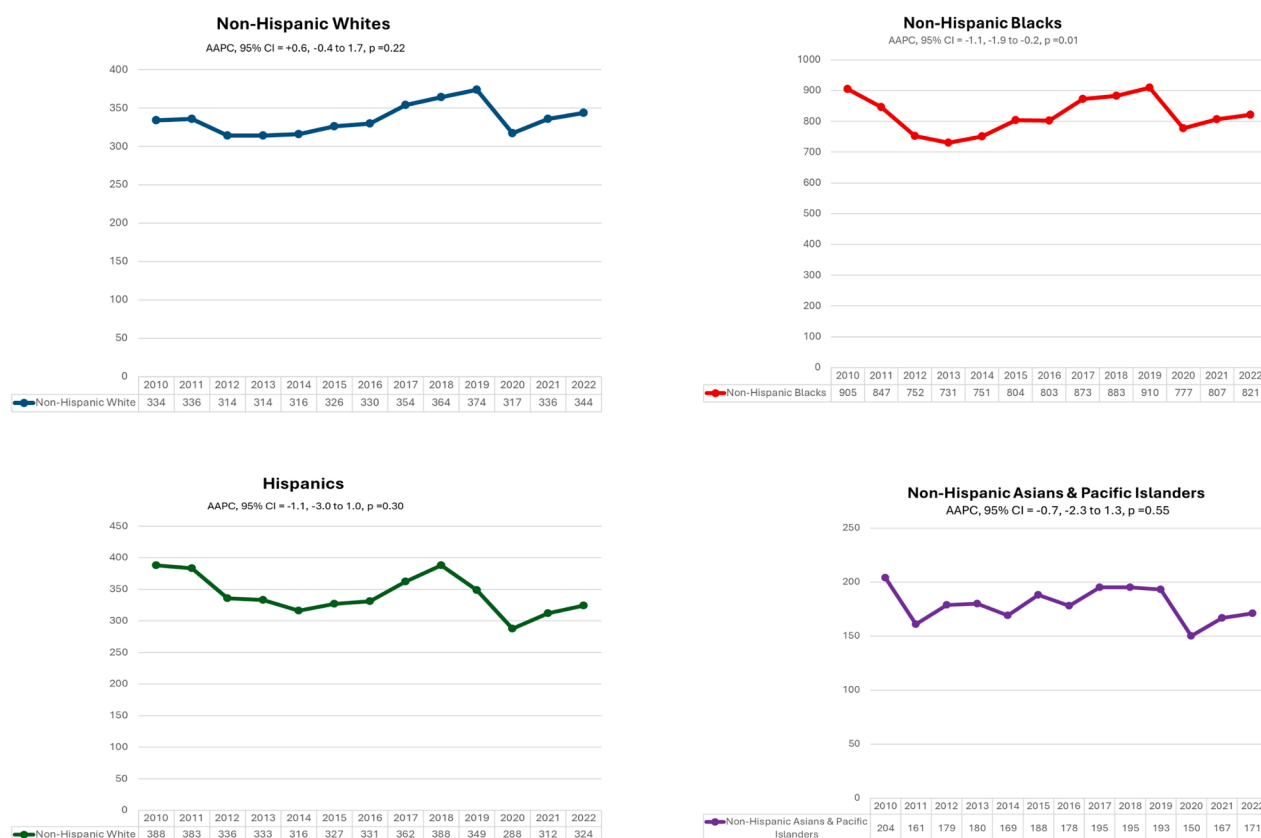
preventive care among racial minorities.¹⁶ HF hospitalizations rates among NH-White adults did not demonstrate a significant reduction from 2010 to 2022. This may reflect a growing burden of high cholesterol, myocardial infarction, angina pectoris,

coronary artery disease, and atrial fibrillation in this population.¹⁷⁻¹⁹ In addition, faster increases in years of potential life lost per 100,000 population for hypertensive heart disease from 2011 onward have been observed for White women

TABLE 2 National Trends in Age-Standardized Primary HF Hospitalization Rates (per 100,000 U.S. Adults) From 2010 to 2022, Stratified by Sex and Age Group

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	AAPC	P Value
Female	349	354	330	323	324	337	341	368	375	381	317	336	332	+0.1 (−1.1 to 1.4)	0.830
Male	480	480	457	455	459	477	479	517	533	541	459	483	464	−0.4 (−1.2 to 0.3)	0.240
Female, ≥65 y	1,588	1,617	1,510	1,463	1,451	1,500	1,498	1,608	1,630	1,641	1,328	1,405	1,379	−0.8 (−2.0 to 0.5)	0.240
Male, ≥65 y	2,047	2,082	1,968	1,947	1,932	1,983	1,953	2,097	2,130	2,139	1,731	1,806	1,712	−1.7 (−2.6 to −0.9)	<0.001
Female, <65 y	93	93	87	88	91	97	102	112	117	121	108	116	116	+1.6 (0.9-2.3)	<0.001
Male, <65 y	156	150	145	148	156	166	174	190	202	210	195	209	207	+2.5 (2.0-3.0)	<0.001

AAPC = average annual percentage change; other abbreviation as in Table 1.

FIGURE 2 Age-Standardized Primary Heart Failure Hospitalization Rates (per 100,000 U.S. Adults)

Age-standardized primary heart failure hospitalization rates (per 100,000 U.S. adults) from 2010 to 2022, stratified by race/ethnicity. Created with BioRender.com.
AAPC = average annual percentage change; NH = non-Hispanic; PI = Pacific Islander.

(mean annual percentage change: 4.2% per year; 95% CI: 0.9%-7.7% per year after 2011) and White men (3.3% per year; 95% CI: 3.0%-3.5% per year after 2011).²⁰

Trends in HF hospitalization rates by insurance payer for the overall cohort showed an increase from 2010 to 2022, except for private insurers. Medicaid beneficiaries experienced a rise compared with private insurers, overlapping with Medicaid expansion.²¹ Hospital location-based analyses revealed declining hospitalization rates in rural centers compared with urban facilities, with an overall decline from 2012 onward. This pattern may be partially explained by the increasing rate of rural hospital closures,²² changes in patient demographics due to rural-urban migration,²³ and transfer protocols facilitating transfers from rural to urban hospitals.²⁴ The barriers to health care access for rural patients, such as cultural attitudes and the absence of health care services, have also been described.²⁵

Beyond differences in hospitalization rates, compared to the 2010-2019 period, age-standardized in-hospital mortality rates had a significant increase in 2020, coinciding with the onset of the COVID-19 pandemic. Among the younger age group, in-hospital mortality remained largely stable, a finding that contrasts with a recent study reporting an increase in overall HF-related mortality in young adults from 2012 to 2019 in the United States.²⁶ This discrepancy likely reflects differences in study design, as Jain et al²⁶ included out-of-hospital events whereas our study captures only in-hospital deaths after hospital admission. Given these distinctions, further research is warranted to examine community-based HF mortality patterns outside hospital systems and the factors contributing to out-of-hospital HF-related deaths. Furthermore, our study expands on prior literature to provide age-standardized trends in in-hospital mortality across sex, race/ethnicity, insurance payer, and urban/rural

TABLE 3 National Trends in Age-Standardized Primary HF Hospitalization Rates (per 100,000 U.S. Adults) From 2010 to 2022 for the Overall Cohort and by Age Groups Based on Race/Ethnicity

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	AAPC	P Value
NH-White	334	336	314	314	316	326	330	354	364	374	317	336	344	+0.6 (−0.4 to 1.7)	0.220
NH-Black	905	847	752	731	751	804	803	873	883	910	777	807	821	−1.1 (−1.9 to −0.24)	0.010
Hispanic	388	383	336	333	316	327	331	362	388	349	288	312	324	−1.1 (−3.0 to 1.0)	0.300
NH-Asian and PI	204	161	179	180	169	188	178	195	195	193	150	167	171	−0.7 (−2.3 to 1.3)	0.550
NH-White, ≥65 y	1,603	1,614	1,492	1,467	1,464	1,498	1,497	1,601	1,617	1,639	1,327	1,407	1,464	+0.6 (−1.7 to 0.6)	0.340
NH-Black, ≥65 y	3,065	2,906	2,527	2,399	2,397	2,550	2,499	2,705	2,717	2,777	2,274	2,314	2,409	−2.5 (−3.5 to −1.5)	<0.001
Hispanic, ≥65 y	1,727	1,764	1,520	1,485	1,400	1,444	1,443	1,546	1,646	1,439	1,163	1,233	1,315	−2.2 (−3.8 to −0.5)	0.009
NH-Asian and PI, ≥65 y	977	757	831	827	754	873	796	848	856	844	634	701	718	−1.7 (−3.9 to 0.9)	0.190
NH-White, <65 y	73	73	72	76	79	85	89	97	106	113	108	114	112	+3.7 (2.9–4.6)	<0.001
NH-Black, <65 y	456	420	383	385	409	441	450	492	501	522	465	492	490	+0.4 (−0.2 to 1.2)	0.180
Hispanic, <65 y	112	98	91	96	93	97	102	117	129	124	107	122	119	+0.7 (−0.5 to 1.8)	0.280
NH-Asian and PI, <65 y	45	37	44	46	47	46	49	59	57	58	49	56	58	+2.8 (1.5–4.2)	<0.001

Abbreviations as in Tables 1 and 2.

hospital locations for younger and older age groups, incorporating data spanning both before and after the onset of the COVID-19 pandemic. The increasing mortality trend after 2019 is similar to the study by Woodruff et al,²⁷ where older adults experienced disproportionately large increases in heart disease mortality rates. This can plausibly be due to COVID-19 spillover effects of disruptions in outpatient care, preventive screenings, and treatment of chronic conditions, further complicated by worsening social determinants of health, such as unemployment and financial hardship.¹³

STUDY LIMITATIONS. The NIS does not track patients across hospitalizations, and thus index hospitalizations and rehospitalizations could not be differentiated. The NIS is an administrative database, and clinical data, such as medications and ejection fraction, are not available. Hence, we were not able to analyze poor prognosis factors, such as lack of guideline-directed medical therapies or ejection fraction, among younger patients.²⁸ We cannot exclude plausible reasons behind the decreasing trend among older adult primary HF hospitalizations due to shifts in administrative coding practices to

TABLE 4 National Trends in Age-Standardized Primary HF Hospitalization Rates (per 100,000 U.S. Adults) From 2010 to 2022 for the Overall Cohort and by Age Groups Based on Rural/Urban Location and Primary Insurance Status

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	AAPC	P Value
Urban	458	454	402	399	411	429	436	473	490	502	427	465	461	−0.2 (−1.1 to 0.7)	0.50
Rural	249	240	258	249	216	216	212	220	212	208	169	183	176	−3.0 (−4.3 to −1.8)	<0.001
Urban, ≥65 y	1,950	1,961	1,723	1,686	1,724	1,790	1,790	1,932	1,977	2,000	1,646	1,814	1,777	−0.1 (−1.9 to 1.8)	0.92
Rural, ≥65 y	1,226	1,167	1,251	1,191	1,020	995	966	1,002	945	914	719	779	734	−4.4 (−5.6 to −3.2)	<0.001
Urban, <65 y	152	145	131	135	142	150	158	174	185	194	177	188	191	+1.9 (1.3–2.5)	<0.001
Rural, <65 y	49	49	54	56	51	56	57	59	62	63	56	61	61	+1.9 (0.7–3.1)	0.002
Medicare	1,016	1,029	948	934	941	975	1,020	1,166	1,218	1,225	1,137	No insurance-based census data available		+0.8 (0.1–1.4)	0.020
Medicaid	346	296	287	285	293	288	301	335	359	400	393		394	+1.7 (1.0–2.4)	<0.001
Private	73	67	58	60	63	65	64	68	72	75	74		76	+0.2 (−0.3 to 0.7)	0.470
Medicare, ≥65 y	1,742	1,767	1,630	1,581	1,564	1,610	1,601	1,714	1,720	1,712	1,505		1,450	−1.6 (−2.3 to −1.1)	<0.001
Medicaid, ≥65 y	239	162	155	157	164	180	175	188	192	200	200		215	+1.8 (1.1–2.6)	<0.001
Private, ≥65 y	197	176	142	150	161	158	155	164	170	181	177		177	−0.7 (−1.5 to 0.7)	0.19
Medicare, <65 y	864	874	804	798	810	842	897	1,051	1,113	1,123	1,059		1,067	+1.6 (0.9–2.3)	<0.001
Medicaid, <65 y	368	325	315	312	320	311	327	366	395	442	434		431	−0.4 (−1.3 to 1.0)	0.490
Private, <65 y	47	44	40	41	43	46	45	48	51	53	52		55	+1.1 (0.4–1.9)	0.002

Abbreviation as in Tables 1 and 2.

FIGURE 3 Age-Standardized and Unadjusted In-Hospital Mortality



(A) Age-standardized in-hospital mortality (%) from 2010 to 2022 for the overall cohort and by age group: younger (<65 years) and older (≥65 years). Age-standardized to the 2000 U.S. standard population. (B) Unadjusted in-hospital mortality (%) from 2010 to 2022 for the overall cohort and by age group: younger (<65 years) and older (≥65 years).

avoid penalties.²⁹ Ethnicity data for Hispanic and NH individuals are not ascertained as a separate variable in the NIS. To address varying rates of missing data on race/ethnicity across the study period (10.2% in 2010 vs 2.1% in 2022), we applied a multinomial logistic imputation model incorporating patient and hospital characteristics to impute race/ethnicity for each year, as recommended by the NIS.³⁰ However, imputations may not fully account for racial/ethnic disparities. In addition, age-standardized rates for

insurance subgroups could not be calculated for 2021 because of the absence of census data for 2021. Our study lacks data to establish causality regarding the COVID-19 effect on trends. Despite these limitations, our study provides contemporary age-standardized hospitalization rates with age-based adjustments for the overall cohort and by subgroups of race/ethnicity, sex, insurance payer, and urban/rural location. The NIS is the largest representative data set for all-payer hospitalizations in the United States.

TABLE 5 National Trends in Age-Standardized In-Hospital Mortality From 2010–2022 for the Overall Cohort and by Age Groups Based on Sex, Race/Ethnicity, Insurance Status, and Rural/Urban Location

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Female	1.6	2.2	1.7	1.5	1.5	1.8	1.8	1.4	2.0	1.4	1.9	1.4	1.8
Male	2.0	1.7	2.1	2.0	2.0	1.9	2.0	2.0	1.9	1.9	2.7	2.2	1.9
NH-White	2.2	2.6	2.2	1.9	1.7	2.6	2.3	2.5	2.0	1.8	2.9	2.0	2.3
NH-Black	1.4	1.4	1.4	1.3	1.3	1.2	1.8	1.1	1.6	1.4	1.5	1.6	1.7
Hispanic	1.2	1.2	1.7	2.9	1.9	1.7	1.2	1.7	1.8	1.9	2.2	1.7	1.5
NH-Asian and PI	5.8	2.0	1.4	2.3	1.3	1.3	1.5	1.5	1.7	1.2	1.6	2.3	2.0
Urban	1.8	1.9	2.0	1.8	1.8	1.9	2.0	1.8	1.9	1.6	2.4	1.9	1.8
Rural	1.6	0.9	1.4	1.2	1.1	1.3	0.9	1.0	0.8	1.8	1.3	1.3	2.1
Medicare	1.7	1.6	1.7	2.0	1.6	1.5	2.1	1.4	1.5	1.4	2.0	1.8	2.0
Medicaid	1.8	1.8	2.4	1.9	1.8	2.0	1.8	1.4	1.9	1.5	2.6	1.8	1.7
Private	2.1	2.3	2.0	2.0	2.0	2.3	1.8	2.7	2.1	1.9	1.9	1.8	1.9
Age <65 y													
Female	1.4	2.2	1.5	1.2	1.3	1.7	1.6	1.1	2.0	1.2	1.7	1.1	1.6
Male	1.7	1.3	1.9	1.7	1.7	1.6	1.8	1.8	1.6	1.7	2.6	1.9	1.5
NH-White	2.0	2.6	2.0	1.6	1.5	2.4	2.2	2.4	1.9	1.6	2.9	1.8	2.0
NH-Black	1.3	1.3	1.3	1.1	1.1	1.1	1.7	0.9	1.5	1.3	1.3	1.5	1.5
Hispanic	0.9	0.9	1.5	3.0	1.7	1.5	1.0	1.6	1.6	1.8	2.1	1.5	1.3
NH-Asian and PI	6.3	1.9	1.1	2.1	0.9	1.0	1.1	1.3	1.4	0.8	1.3	2.1	1.7
Medicare	1.5	1.4	1.5	2.1	1.5	1.3	1.7	1.1	1.3	1.3	1.9	1.7	1.8
Medicaid	1.7	1.8	1.8	1.6	1.3	1.3	1.4	1.3	1.7	1.3	1.6	1.3	1.4
Private	1.7	1.6	1.5	1.6	1.2	1.9	1.5	1.8	1.5	1.5	1.5	1.6	1.6
Urban	1.6	1.7	1.8	1.6	1.6	1.7	1.9	1.5	1.8	1.4	1.9	1.6	1.5
Rural	1.4	0.6	1.3	0.9	0.6	1.0	0.6	0.8	0.5	2.0	0.9	1.0	2.0
Age ≥65 y													
Female	2.8	2.7	2.7	2.7	2.7	2.7	2.7	2.6	2.5	2.3	2.8	2.9	2.9
Male	3.4	3.3	3.4	3.4	3.3	3.3	3.1	3.2	3.0	2.9	3.2	3.5	3.5
NH-White	3.3	3.3	3.2	3.3	3.2	3.2	3.1	3.1	2.9	2.8	3.1	3.4	3.4
NH-Black	2.3	2.0	2.3	2.3	2.2	2.1	2.2	2.2	2.1	1.9	2.5	2.5	2.7
Hispanic	2.4	2.6	2.4	2.6	2.6	2.8	2.6	2.5	2.4	2.4	2.7	2.8	2.4
NH-Asian and PI	2.8	3.1	2.7	3.1	2.8	2.6	3.1	2.4	2.9	3.0	2.9	3.4	3.3
Medicare	3.2	3.0	3.1	3.1	3.0	3.0	2.8	2.9	2.7	2.6	3.0	3.2	3.1
Medicaid	2.4	2.2	2.7	2.7	2.3	2.8	2.0	2.1	2.5	2.0	2.7	3.1	2.9
Private	4.1	4.8	5.2	5.0	4.9	5.0	5.0	4.8	4.1	4.1	3.7	3.8	3.4
Urban	3.1	3.0	3.0	3.1	3.0	3.0	2.9	2.9	2.8	2.7	3.0	3.2	3.2
Rural	3.1	2.9	3.1	2.9	2.9	2.8	2.7	2.7	2.4	2.4	2.8	3.0	3.1

Values are %.
Abbreviations as in Table 1.

Hence, despite the limitations, the NIS remains the most comprehensive database to examine long-term trends in hospitalizations in the United States.

CONCLUSIONS

In this longitudinal analysis of U.S. hospitalizations from 2010 to 2022, we examined age-standardized rates of primary HF hospitalizations and in-hospital mortality. HF hospitalization rates increased from 2014, peaking in 2019, followed by a sharp decline in 2020, which persisted through 2022, with a more pronounced decline among

adults aged ≥65 years. Notably, we observed a sustained upward trajectory in HF hospitalization rates among younger adults across the study period, reflecting a shift in the epidemiological landscape in which HF hospitalization is increasingly affecting younger populations. Age-standardized in-hospital mortality rates have risen since 2020, disrupting trends from prior years. Such findings highlight the need for further research to better understand the factors contributing to the observed trends, which will be essential in guiding interventions targeted at improving outcomes for patients with HF.

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PERSPECTIVES

COMPETENCY IN SYSTEMS-BASED PRACTICE:

HF hospitalization rates increased from 2014 onward, followed by a sharp decline in 2020, with a more pronounced decline among older adults (≥ 65 years). There was an upward trajectory in HF hospitalization rates among younger adults (< 65 years), and rates decreased at rural hospitals. Age-standardized in-hospital mortality increased in 2020, disrupting trends from prior years.

TRANSLATIONAL OUTLOOK: These observations highlight the need for further research to better understand factors that are essential to guide interventions targeted at improving outcomes for patients with HF.

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KEY WORDS demographics, HCUP, heart failure, outcomes, trends

APPENDIX For supplemental tables and figures, please see the online version of this paper.